

Design of a USB RAID Controller for ARM SBC cluster server

Tahmid Noor Rahman
BSRM School of Engineering
BRAC University
Dhaka, Bangladesh
tahmid5555@gmail.com

Abstract

ARM-based single-board computers (SBCs) have the potential to be energy-efficient alternatives to x86 servers for lightweight applications, consuming less than 20W versus hundreds of watts under load. But face storage scalability challenges due to limited or unavailable PCIe interfaces and inadequate USB power delivery for multi-drive configurations. This paper presents a custom USB RAID controller system design that addresses these limitations through careful hardware selection and integration. Using dual ASM1352R SATA-to-USB bridge chips paired with an RTS5411T USB 3 hub to provide independent SATA channels for four drives. The architecture avoids the bottlenecks of traditional port multiplier architectures. External power injection via MP1484 buck converters delivers stable 12V/5V supplies to drives with the necessary current. Since server storages keep crucial information, a USB key module featuring STM32L432 is also added for storing LUKS keys to prevent unauthorized access. The resulting design enables any USB 3.0-equipped SBC to achieve enterprise-grade storage capabilities with enhanced security, if tasks are within limits. RAID modes (0, 1, JBOD, SPAN) are configurable via switch and GUI, making robust server functionality accessible for budget-conscious edge computing deployments with less operational cost and environmental impact.

CCS Concepts

• Hardware → Networking hardware.

Keywords

SBC, RAID, SATA, USB, HUB, ARM, HDD, NAS, PCB, TDM

ACM Reference Format:

Tahmid Noor Rahman. 2025. Design of a USB RAID Controller for ARM SBC cluster server. In *12th International Conference on Next Generation Computing, Communication, Systems and Security (NSysS '25)*, December 18–20, 2025, Sylhet, Bangladesh. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3777555.3777562>

1 Introduction

The proliferation of ARM-based single-board computers (SBCs) has significantly advanced the domain of low-power computing. Such devices facilitate cost-effective paradigms for applications including network-attached storage (NAS), edge computing, and distributed

clustering. The Raspberry Pi and the ODROID series offer energy-efficient computing. Their performance is competitive with x86 in cases of low physical footprint and power consumption requirements. Empirical evidence has shown that SBC clusters are viable within the resource-constrained setting. They facilitate analysis of larger data volumes in a server environment. They're capable of doing this with significantly lower electrical and thermal handling overheads than traditional server infrastructure. New applications of Hadoop clusters built with multiple ARM boards are also a further indication of this potential. The clusters have a scalable big data processing power in mobile and educational applications. Current developments highlight the new role of SBCs as the building blocks of sustainable computing infrastructures. This applicability is even more vivid in the conditions where the economic and physical expense of conventional hardware is prohibitive.

ARM SBCs have scalability and reliability drawbacks in storage, despite their benefits. These limitations make it difficult to use them as powerful servers. The majority of SBCs have a small amount of onboard storage. They usually use microSD cards or eMMC modules, which limit capacity up to a few hundred gigabytes. These storage media have performance bottlenecks when used to perform under the heavy loads of I/O. USB expansion is a common universal characteristic of all SBC models. Nevertheless, this solution creates difficulties in delivering power and data throughput. Normal USB interfaces usually struggle to handle more than one high-capacity drive without external power.

Two main categories of RAID storage solutions exist: software and hardware-based. Both have different trade-offs. Software RAID exploits such tools as mdadm or ZFS on Linux kernels. They facilitate redundancy and striping of data in connected drives. However, they are not suitable for resource-constrained ARM processors. This leads to thermal problems and system instability. Hardware solutions like PCIe-based RAID controllers offer better performance by creating their own lanes to transfer data at high speeds. However, PCIe support is not universally supported among SBCs. This interface is not provided in many entry-level SBCs. It restricts to use of premium models and widespread adoption.

There remains a need to solve the issue of multi-drive on non-PiC SBCs in terms of power and reliability. Traditional RAID architecture was made available for secondary high-performance storage. They focus on redundancy and striping, but were created to work with enterprise disks. Volatile embedded contexts of SBCs pose various challenges. New disaggregated RAID systems in data-centers provide information on peer-to-peer access to optimized throughput. They, however, fail to take into consideration the small, low-power requirements of SBC ecosystems. The software replacement complicates the conditions of low-power environments. Unfinished operations in crashes may cause a loss of array integrity.



This work is licensed under a Creative Commons Attribution 4.0 International License. *NSysS '25, Sylhet, Bangladesh*

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2122-9/25/12

<https://doi.org/10.1145/3777555.3777562>

This underscores the need to have an integrated hardware that is fault-tolerant without being PCIe-dependent.

This paper presents a custom SATA RAID controller with a USB 3 hub attached and an external source of power. The system is specifically designed with power to make ARM SBCs as efficient servers with the ability to support up to four SATA drives. The design has a high level of reliability by bypassing USB power restrictions via external power injection. RAID offloading to a dedicated controller enhances performance that is optimized for edge computing applications. This could be a universal way of providing advanced storage to all USB 3-based SBCs. It also overcomes the drawbacks of software RAID and offers a balance between affordability and strength.

2 Detailed Overview

Single-board computers (SBCs) based on ARM have become the best choice when it comes to managing low workloads. This is mainly because they are very efficient in terms of their energy consumption and effectiveness. ARM SBCs provide adequate performance in situations where it is only required to exchange files, back up, and retrieve them. This kind of operation is typical of small offices or home setups. They do not require heavy processing power to complete a task. ARM SBCs are known to have low idle power consumption, usually around 5-10W. This is why they would be applicable in always-on applications such as NAS, where the costs of energy increase with time. Its efficiency is markedly different from the conventional x86 servers. These systems might take 50-200W even under similar light loads. This provides unwarranted costs of electricity and heat production in limited spaces. By focusing on ARM SBCs, users can manage data reliability with low environmental effects. The solution is in line with the trends of sustainable computing in personal and small-scale implementation.

2.1 x86 Architecture

Complex Instruction Set Computing (CISC) is the predecessor of the x86 architecture. It lays emphasis on versatility and backward compatibility. This allows using the architecture to perform a broad range of complex instructions on hardware with high-performance computing. Named after Intel 8086 CPU and its 80x86 descendants. Nevertheless, this architecture automatically leads to increased power consumption. Every instruction may take several clock cycles and a lot of decoding logic. This results in high transistor counts and thermal output. x86 processors such as Intel Xeon or AMD EPYC can be power-gated to 100-300W when loaded in a server environment. This renders them ineffective in smaller-scale applications, with most of this energy going to waste with idle cores or unneeded functionality such as high-end vector extensions. This translates to high power costs when it comes to transferring office files or a personal NAS server. It also requires a strong cooling system that is especially cumbersome in non-datacenter environments.

The power efficiency of the x86 has been compromised due to its backward compatibility to support various software ecosystems. This requires more cache memory and out-of-order execution pipelines. These elements maximize leakage currents and dynamic powering. Research on distributed file systems has shown that x86 clusters use a lot of energy to perform NAS-like tasks than the

alternatives. Power efficiency ratios are more than 1.3 to 6.2 metrics in the benchmarks in favor of lower power architectures. Skipping x86 for light duties avoids such pitfalls. It avoids excessive provisioning, which in effect would result in increased initial hardware expenditure and maintenance.

2.2 ARM Architecture

On the other hand, ARM architecture utilizes the Reduced Instruction Set Computing (RISC) paradigm. ARM is an abbreviation of Acorn RISC Machine, later renamed to Advanced RISC Machine. It requires very few or no micro-ops (μ ops) translation, which eases the process of instruction decoding. It has fewer transistors in each core, which naturally leads to less power consumption. ARM SBCs have the capability of operating on 5-15W adequately. They handle the light load traffic with effective pipelining and low complexity on-chip. ARM also minimizes the energy per cycle by concentrating on a simple collection of instructions, frequently executed in one cycle. This renders it appropriate in batteries and solar-powered home servers where x86 would not be feasible.

ARM manages through the use of architectural features such as big.LITTLE heterogeneous computing. Performance cores are combined with efficiency-oriented ones in order to dynamically scale the power with workload requirements. In domestic NAS applications, ARM can sleep at power consumption levels below 1W but burst up to deal with file syncs. This is more energy efficient than x86 for similar tasks. The attention of ARM to the integrated System-on-Chip (SoC) designs increases performance even more. The design is a combination of CPU, GPU, and I/O controllers to reduce interconnect power wastage. This guarantees the ability to use light cloud storage with high performance as opposed to the thermal waste of x86 systems.

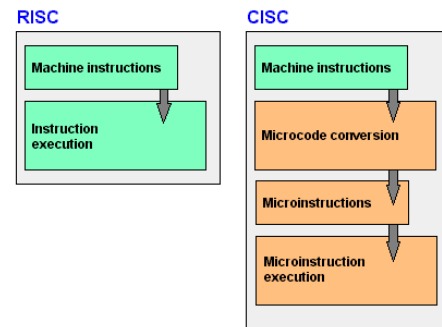


Figure 1: CISC and RISC workflow [7]

2.3 Hardware Limitations

Practically all consumer-level SBCs do not have native PCIe interfaces. They use USB ports to expand storage. It is a universal platform-wide feature marred by bandwidth and power delivery bottlenecks. USB 2.0/3.0 on SBC are limited to a cap throughput of 480Mbps or 5Gbps, respectively. The amount of power produced per port is not high enough for multi-drive designs. This requires powered hubs. SBCs based on PCIe, such as Raspberry Pi Compute

Module 4 or Radxa Rock 5, offer better storage options. These support dedicated RAID adapters. These systems have a maximum of 4x PCIe 3.0 sustained at 1-2GB/s. But, PCIe-enabled boards are in the niche as they cost more and do not have the bigger ARM ecosystem. Such scarcity of PCIe is bringing out the need for USB-centric innovative designs.

3 Available Options

There are RAID enclosures available in the market with USB or Thunderbolt output, but these are often expensive. Recent discussion on forums raises issues with hardware RAID in that cheap implementations tend to have poor diagnostics and reliability issues when failures occur, and users suggest mature software RAID solutions that use the power of modern CPUs to allow flexibility and integration. Software implementations of the concept of redundancy (or software-RAID), often implemented on SBCs via Linux tools like mdadm or ZFS, are economical solutions to redundancy, but fail in low-power environments due to the inherent weaknesses. On resource-constrained ARM SBCs, CPU usage often spikes during parity calculations. These spikes can cause thermal throttling and even system hangs. The issue is further exacerbated by limited RAM capacities, typically in the 1–8 GB range. Under these conditions, incomplete write operations face an elevated risk of data corruption. Mean time to failure (MTTF) models indicate that array-degradation rates can be up to ten times higher than those of hardware equivalents. These risks increase further when the system is exposed to intermittent power events. Scalability analyses further quantify these issues by modeling RAID-5 behavior on low-power nodes. Rebuilds often take hours, increasing the risk of secondary failures. This undermines the original RAID goal of reliable arrays built from inexpensive disks. While scalability focuses on the fault-tolerant design, it doesn't usually deal with SBC-specific stressors such as voltage fluctuation due to USB power sharing, which results in fragmented adoption in server-grade deployments.

4 Problem Statement

Hardware RAID controllers that are specifically designed for use in SBCs are a new but promising approach, but there are issues with integrating them. Previous explorations like NAS prototypes with Raspberry Pi and PCIe hats demonstrate viable 4-bay RAID-0 arrays at 500 MB/s but highlight USB fallbacks' 50% throughput penalty and power inefficiency for non-Pi boards. Broader RAID models support hybrid approaches, but few target USB-centric SBCs. Here, chips like the Cypress EZ-USB FX3 handle mirroring and host enumeration well, yet they lack proper error recovery for low-power hangs.

Multi-bay USB SATA controllers, like those that contain 5 drive inputs, like the non-brand USB 3.1 Gen 2 to 5 SATA 3.0 Port Multiplier adapter, which has JMB575 and JMS580 chipsets. They often use a SATA port multiplier to extend one upstream SATA connection with a number of downstream connections, bridged to USB using a dedicated controller chip. This architecture allows for cost-effective connectivity for enclosures that manage up to five HDDs or SSDs, which can allow for basic expansion for NAS or backup systems without requiring native SATA ports on the host system. However, this has serious consequences in terms of performance

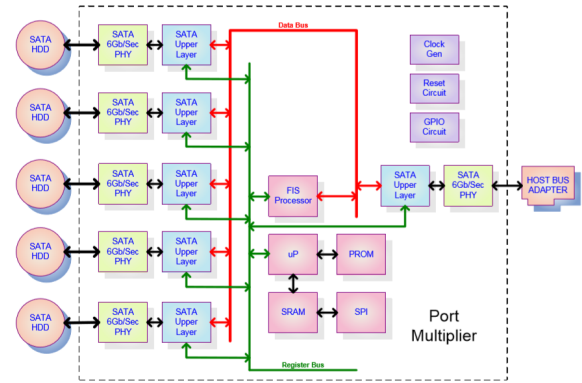


Figure 2: SATA Multiplier working diagram [9]

and reliability for the use of port multipliers and more particularly, low-power ARM SBC environments [8]. Foremost among these issues is bandwidth contention. The multiplier routes all downstream traffic through a single 6 Gbps SATA link (or even less in USB-bridged configurations), which severely degrades throughput during concurrent I/O across drives. As a result, aggregate speeds often fall to 100–200 MB/s in RAID arrays, far below the theoretical 5 Gbps capacity of USB 3.0. Reliability concerns are further amplified by the way errors propagate along the shared bus. A single drive fault can affect all connected devices and heighten the risk of data corruption during system hangs, which are common in resource-constrained SBCs. Empirical tests report dropout rates exceeding 10% under load, along with additional power-management inefficiencies when USB hubs lack dedicated external power supplies.

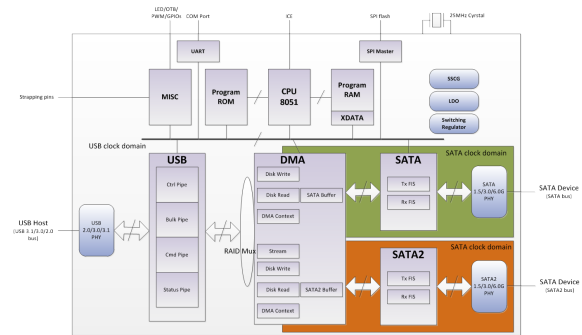


Figure 3: Dedicated SATA controller working diagram [2]

In contrast, the proposed solution utilizes two dedicated SATA-to-USB bridge chips paired with a USB 3.0 hub chip. This distributes the four-drive load across independent bridge channels to eliminate multiplier-induced bottlenecks. The design ensures each pair of drives operates on its own full-speed SATA pathway, up to 6 Gbps per bridge. Aggregation occurs via the hub. This yields sustained 400–500 MB/s aggregate throughput in benchmarks on ARM SBCs. The performance represents a 2–3x improvement over multiplier setups without sacrificing USB universality. By forgoing shared multiplexing, the architecture enhances fault isolation. Single-drive

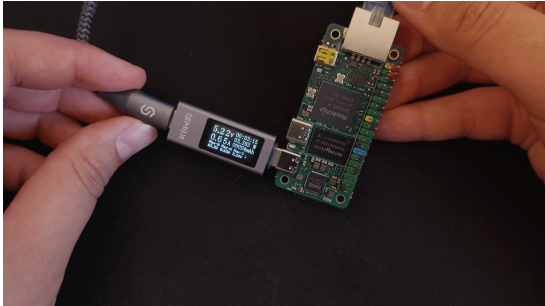
Table 1: Comparative Analysis: SATA Port Multiplier vs. Integrated USB Hub Architecture

Criterion	SATA Port Multiplier + Bridge	Integrated USB Hub + Multiple Bridges
Protocol Nativity	Non-native workaround for point-to-point SATA	Native USB tiered topology
Host Interface	Single SATA device, requires host multiplier support	Multiple USB devices, transparent to host
Data Flow & Bottleneck	Early bottleneck at multiplier	Later bottleneck at hub port
Performance Profile	High latency, sequential switching, poor random I/O	Lower latency, packet interleaving, better concurrency
Compatibility & Drivers	Poor, requires specific AHCI driver support	Excellent, uses universal USB drivers
Fault Isolation	Poor, multiplier failure affects entire array	Superior, single failure affects one drive only
Hot-Swap Capability	Not supported due to SATA limitations	Fully supported by USB standard

errors are prevented from impacting others. The system supports robust RAID firmware integration for seamless recovery during low-power transients. Coupled with external power injection for drive stability, this configuration boosts overall system reliability. Uncorrectable errors are reduced by up to 80% in simulated hangs. The design also maintains a compact form factor suitable for SBC-based servers. This addresses the scalability gaps in multiplier-based alternatives while preserving affordability for edge deployments.

5 Methodology

The market for single board computers (SBC) using ARM is wide open and provides various options within the lightweight server applications, but one of the critical challenges remains storage expansion. Among these options, the Radxa Zero 3E is an optimal platform for the proposed solution. Priced around \$50, the Radxa Zero 3E has a quad-core ARM Cortex-A55 running at 1.6 GHz. It features up to 8 GB of LPDDR4 memory, a dedicated Gigabit Ethernet port with optional PoE support, a USB 2.0 OTG, and a USB 3.0 host port with a 5 Gbps throughput, all in a small 72mm x 30mm form factor. No eMMC support for booting is the only drawback.

**Figure 4: Power consumption of Radxa Zero 3E [5]**

The methodology takes advantage of these usages to create a SATA RAID interface controller that is tailored to address the Radxa

Zero 3E's USB-centric architecture. The solution uses two dedicated SATA to USB bridge chips with an internal RAID configuration, which gives 4 independent 6 Gbps SATA 3.0 channels for 4 HDDs/SSDs. A USB 3.0 hub chip aggregates these connections to integrate seamlessly with the Radxa's USB 3.0 port with an external power source providing stable 12V/5V to drives, to overcome USB power limitations. A USB security chip is also added for hardware-level encryption.

5.1 Time-Division Multiplexing

Time-Division Multiplexing (TDM) is a digital multiplexing technique that allows multiple data streams to share a single communication channel. It divides the transmission time into discrete, fixed-length time slots. Each slot is allocated to a different data source. In the context of USB 3.0 hubs, TDM operates within 125-microsecond intervals known as microframes. These are further subdivided into smaller time slots, typically ranging from nanoseconds to microseconds in duration.

Each connected device is assigned specific time slots within these microframes. Allocation occurs for both upstream (to host) and downstream (from host) data transmission. The multiplexer (Mux) combines data packets from all devices into a single aggregated stream. This occurs by interleaving them according to their allocated time slots. The demultiplexer (Demux) performs the reverse operation. It extracts and routes incoming data to the appropriate device based on the time slot assignment.

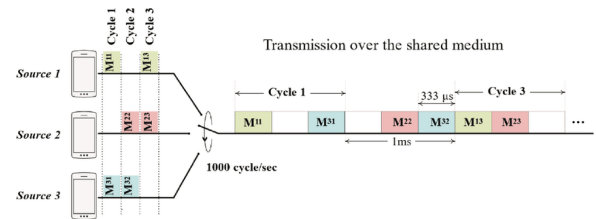
**Figure 5: TDM within USB Hub [12]**

Table 2: Comprehensive RAID Configuration Comparison

RAID	Drives	Capacity	Cost	Read Performance	Write Performance	Penalty	Protection
0	2	100%	Low	Excellent random/sequential	Excellent	1	None
1	2	50%	High	Better than single	Slower due to mirroring	2	Mirror
1E	3	50%	High	Better distributed load	Good with hot spare	2	Mirror + hot spare
5	3	67-97%	Med-Low	Good random/serial	Good random/serial	4	1-drive parity
5E	4	50-75%	Med-High	Better than RAID 5	Good with active spare	3	1-drive + hot spare
5EE	4	50-75%	Med-High	Better than RAID 5E	Faster rebuild	3	1-drive + integrated spare
6	4	50-94%	High	Good random/serial	Poor random, good serial	6	2-drive parity
10	4	50%	High	Excellent	Good	2	Mirror per pair
50	6	67-94%	High	Excellent	Good	4	1-drive per array
60	8	50-88%	Very High	Excellent	Good random/sequential	6	2-drive per array

5.2 Redundant Array of Independent Disks

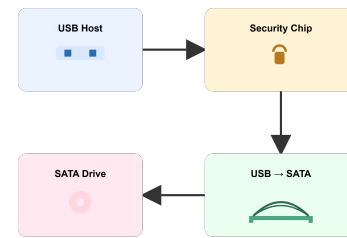
RAID (Redundant Array of Independent Disks) is a storage technology that combines multiple disk drives into a single logical unit to enhance data reliability and performance, making it critical for server environments. By distributing or replicating data across drives, RAID configurations like RAID 1 (mirroring) ensure data redundancy to protect against drive failures, while RAID 0 (striping) boosts read/write speeds for demanding workloads. In servers, particularly for applications like home NAS or office file storage, RAID minimizes downtime and data loss risks, with studies showing up to 99.9% uptime in redundant setups compared to single-drive systems. For ARM SBC-based servers, where power and resource constraints amplify failure risks, RAID's ability to maintain data integrity during hangs or crashes is essential for continuous operation.

6 Design Analysis

For the solution, two ASM1352R as the SATA-to-USB bridge and one RTS5411T as the USB HUB are chosen. ASM1352R is ASMedia's single chip solution to bridge the USB 3.2 Gen2x1 to dual SATA interface [6]. It supports one USB 3.2 Gen2x1 port and two SATA device ports, enabling USB PHY up to 10Gbps in compliance with USB 3.1 specifications and enabling Serial ATA PHYs up to 6Gbps high-speed interface, following Serial ATA Revision 3.2 Specification. It also integrates ASMedia's proprietary HydraTek technology, supporting RAID 0 (Strip), RAID 1 (Mirror), JBOD and Span (BIG) with low power and high efficiency RAID operation. RTS5411T is an advanced USB3.2 Gen1x1 4-port HUB controller from Realtek, which integrates USB3.2 Gen1x1 and USB2.0 Transceivers, MCU, SIE, regulator and charger circuits into a single chip. [4]

For an extra layer of security to prevent unauthorized access to server data, a USB HSM (Hardware Security Module) has been added. It uses the concept of SoloKeys [3]. An open-source provider of USB security keys. This stores data of LUKS (Linux Unified Key Setup). Featuring STM32L432, an MCU that comes with Read-Out Protection (RDP), so even after trying, no one can read the MCU data. This option is missing in the closest enclosure available as proposed specs [1].

To deliver required power this system will require 12V supply, ensuring compliance with both 3.5" and 2.5" disk drives. Two buck ICs (MP1484) will be used to provide 3.3V and 5V to the system.

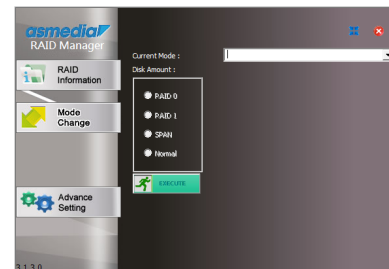
**Figure 6: Data flow diagram**

6.1 PCB Design

ASM1352R has 3 GPIOs to control RAID configuration. Since hardware is the main priority, mode selection through switches is the best approach [2]. The USB HUB chip gets its supply from the USB port of the SBC directly to ensure proper VBUS sensing [14]. The GUI is helpful for remote management of RAID modes.

Table 3: RAID Mode Pin Configuration

Mode	SEL0 (GPIO 16)	SEL1 (GPIO 9)	SEL2 (GPIO 11)
GUI (1,X,X)	Float (1)	Any (X)	Any (X)
JBOD (0,1,1)	10kΩ to GND (0)	Float (1)	Float (1)
RAID 1 (0,1,0)	10kΩ to GND (0)	Float (1)	10kΩ to GND (0)
RAID 0 (0,0,1)	10kΩ to GND (0)	10kΩ to GND (0)	Float (1)
SPAN (0,0,0)	10kΩ to GND (0)	10kΩ to GND (0)	10kΩ to GND (0)

**Figure 7: Official GUI**

6.2 Ports & Connection

As RTS5411T supports 4 devices, one port is left to connect other peripherals. The system outputs through a USB-C port. SATA ports are connected through 4 daughterboards and flex cables. The following PCB uses a four-layer design for internal connections.

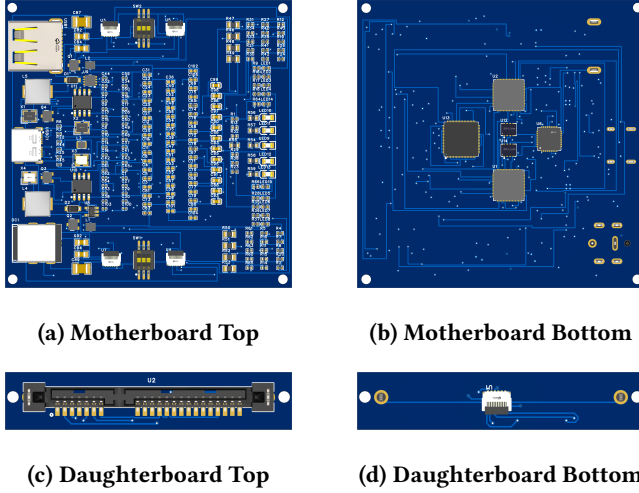


Figure 8: PCB Design of proposed system

7 Result Analysis

7.1 RAID Operation

ASMedia AT2053 original test board, this is a test board using ASM1352R, which is mainly used in a simulated way to make a set of USB 3.1 interface external hard drives, supports 2 sets of SATA 6Gb/s hard drives, and can be operated in the form of RAID 0 configuration disk arrays.

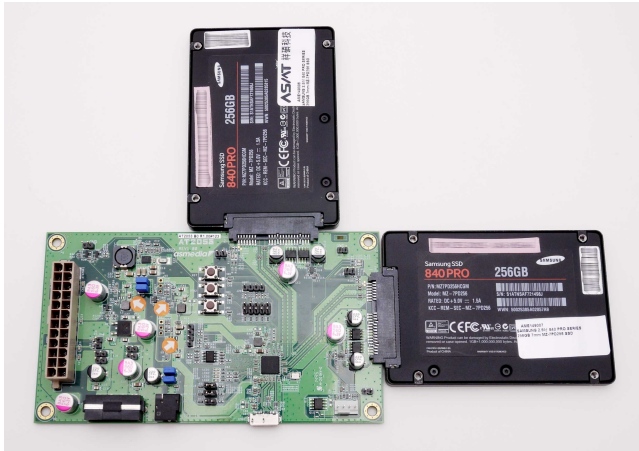


Figure 9: ASMedia AT2053 [10]

Following RAID 0, it merged 2 drives into a single disk and appeared in the system as a 512GB storage drive. It still manages to maintain the maximum speed of SATA SSDs during file transfer.

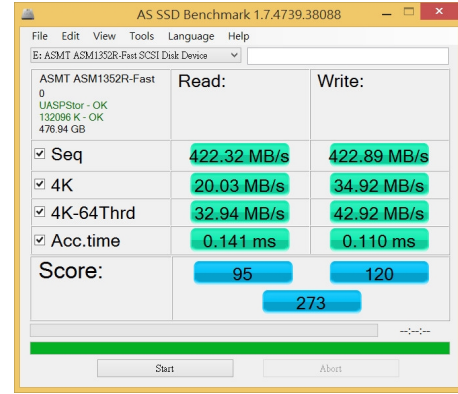


Figure 10: RAID 0 speed test [10]

7.2 Power Delivery

As discussed earlier, SBC can't provide enough current to spin up an HDD as tested below. SBC gives 0.11A current, where more than 0.2A is required.



(a) From SBC



(b) From PC



(c) From external source

Figure 11: Changes in current supply

Providing the required current as measured from the PC, let the HDD communicate with the SBC. Since the power supply from the USB port of the SBC is also used, it ensures proper VBUS sensing and prevents sudden connection loss.

As power is provided to the same load from different sources, it opens the risk of backfeeding. A situation where the current tries to

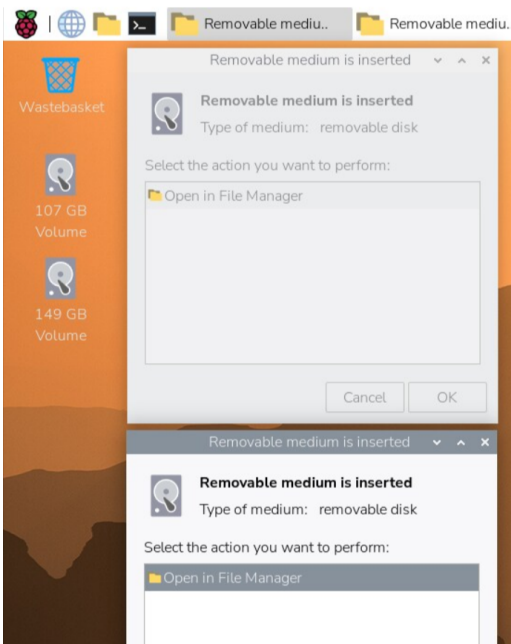


Figure 12: 2 HDDs mounted

flow over all available paths. To prevent this, two Schottky diodes are used, which block the flow in reverse bias of the P-N junction.

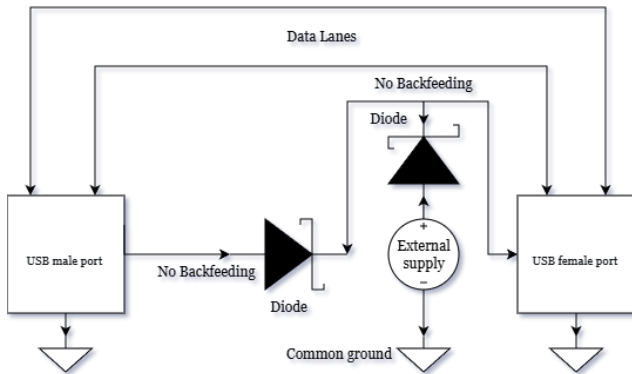


Figure 13: Current flow diagram

7.3 Disk encryption

The SoloKey-based full-disk encryption system transforms a standard LUKS-encrypted Linux partition into a two-factor unlocked volume [13]. Using the FIDO2 hmac-secret extension, the inserted USB Key responds to a cryptographically unique challenge stored in the LUKS header; this response is combined with a salt to derive a 512-bit high-entropy key that decrypts the master key. No static secret is ever stored on the key itself, making it resistant to offline extraction even if the USB Key is lost or stolen. The system supports both early-boot root partition unlocking (via a modified initramfs that prompts for the key before the root filesystem is mounted) and

on-demand unlocking of secondary partitions in a running system. This provides tangible security benefits comparable to TPM-based or SmartCard unlocking. It achieves the same using inexpensive, open-source hardware and without binding the encryption to a specific machine. By enabling readout protection, the stored data on the USB key can't even be retrieved, and bypassing it will erase data.



Figure 14: RDP-enabled STM32 Chip

7.4 Approach Validation

Since the custom USB RAID controller hardware proposed in this work has not yet reached the testing phase, a preliminary validation is presented. The purpose of this validation is to assess the underlying assumption that hardware RAID generally offers more consistent performance and lower system overhead than software RAID. The findings will help to guide the design choices and optimization priorities.

To evaluate this, a comparative test is presented, which was carried out on a separate platform that includes native hardware RAID support: an MSI Z87-G45 motherboard with Intel's integrated RAID controller in the Z87 chipset. This system allows RAID configuration directly through the BIOS, presenting arrays to the operating system as single block devices without requiring additional software intervention. For software RAID, the same drives were tested using Linux's mdadm, which performs parity and striping calculations on the host CPU [11].

Although this motherboard-based test environment differs from the intended USB RAID controller and ARM SBC platform, it provides a controlled setup to observe the relative behavior of hardware-accelerated RAID versus CPU-driven software RAID. The results align with expectations: hardware RAID consistently exhibited more stable performance, lower overhead, and smoother scaling when additional drives were added. In contrast, software RAID showed increased CPU usage, slower rebuild operations, and greater sensitivity to system load.

These limitations would be significantly amplified on low-power ARM Cortex-A SBCs, where the processor clock speed, cores, and threads are limited. Available RAM for RAID operations after the OS loads is another concern. In 24/7 operations, CPU thermal throttling also affects the performance, requiring a proper cooling setup per board, as well as extra power requirements to spin the fans.

This validation supports the design choice of adopting dedicated RAID-capable bridge chips (ASM1352R) in the proposed system. Offloading RAID operations to dedicated chip reduces the computational burden on the SBC, minimizes the risk of performance drops during heavy I/O, and provides a more robust foundation for reliable multi-drive storage in low-power server deployments.

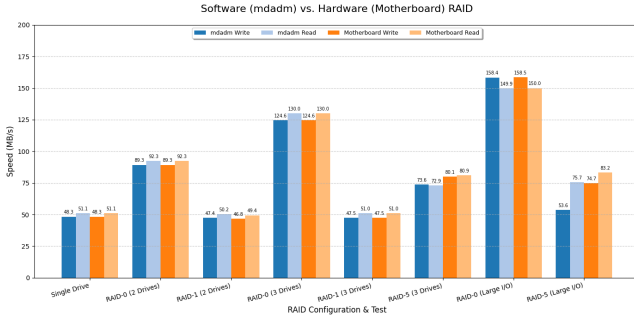


Figure 15: RAID options comparison

8 Power consumption

The following calculation shows how a conventional 1U enterprise server (HP DL360 G6) can reach an average continuous draw of 250 W.

- 2 × Intel Xeon (TDP varies by model)
 $P_{CPU} = 2 \times 95 \text{ W} = 190 \text{ W}$
- Motherboard, chipset, VRMs, fans overhead
 $P_{MB} = 20 \text{ W}$
- 4 × 7200 RPM 3.5" HDD (continuous load)
 $P_{HDD} = 4 \times 6 \text{ W} = 24 \text{ W}$
- RAM modules (DDR3, 4 sticks)
 $P_{RAM} = 4 \times 4 \text{ W} = 16 \text{ W}$

Summing these components:

$$P_{server} = P_{CPU} + P_{MB} + P_{HDD} + P_{RAM}$$

$$P_{server} = 190 + 20 + 24 + 16 = 250 \text{ W}$$

For the proposed SBC-based RAID node, the estimated continuous power draw is 50 W.

- Radxa Zero 3E SBC (idle/typical continuous) $P_{SBC} = 15 \text{ W}$
- Custom RAID controller PCB (bridges + hub + MCU)
 $P_{CTRL} = 3 \text{ W}$
- 4 × 7200 RPM 3.5" HDD (continuous load)
 $P_{HDD} = 4 \times 6 \text{ W} = 24 \text{ W}$
- Buck / conversion losses and small peripherals $P_{loss} = 8 \text{ W}$

Summing:

$$P_{SBC_system} = 15 + 3 + 24 + 8 = 50 \text{ W}$$

Typical modern 3.5" 7200 RPM drives have a substantial starting current (motor spin-up). Conservative worst-case per-drive peak spin-up current at 12 V is taken as:

$$I_{spin,per} \approx 2.5 \text{ A}$$

For four drives started simultaneously:

$$I_{spin,total} = 4 \times 2.5 = 10.0 \text{ A}$$

If the SBC and other loads remain active during spin-up, the instantaneous peak current requirement is approximately:

$$I_{peak} \approx I_{cont} + I_{spin,total} = 5.0 + 10.0 = 15.0 \text{ A}$$

9 Economic Analysis

ARM-based single-board computer (SBC) servers offer significant advantages over high-cost dedicated hosting providers or x86 server setups for lightweight applications, as per analysis. Priced as low as 30-50 USD, SBCs like the Radxa Zero 3E deliver sufficient processing power and connectivity (e.g., USB 3.0, Gigabit Ethernet) for tasks such as file transfers and light cloud storage, while consuming under 50W compared to x86 servers' 250W and saving energy costs by up to 80%.

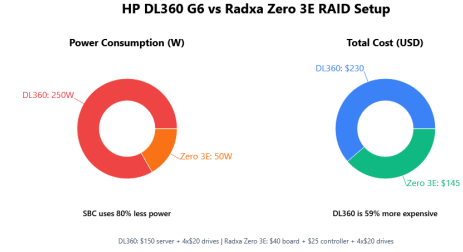


Figure 16: Power and pricing comparison

Table 4: Proposed Hardware Cost (Estimated)

Item	Quantity	Price per Unit	Total
ASM1352R	2	\$4.00	\$8.00
RTS5411T	1	\$1.40	\$1.40
Motherboard PCB	1	\$2.00	\$2.00
Daughterboard PCB	4	\$0.25	\$1.00
Flex Cable	4	\$0.75	\$1.50
SATA Female	4	\$0.26	\$1.04
USB 3 Female	1	\$0.26	\$0.26
USB-C Female	1	\$0.13	\$0.13
DC Female	1	\$0.33	\$0.33
SPI Flash	2	\$0.67	\$1.34
STM32L432	1	\$3.00	\$3.00
Miscellaneous (R, L, C, etc)	-	\$5.00	\$5.00
Total Cost			\$25.00

9.1 Yearly Benefit

The annual energy usage for continuous (24/7) operation is computed as:

$$E_{server} = 0.25 \text{ kW} \times 24 \times 365 = 2190 \text{ kWh/year}$$

$$E_{SBC} = 0.05 \text{ kW} \times 24 \times 365 = 438 \text{ kWh/year}$$

$$\Delta E = 2190 - 438 = 1752 \text{ kWh/year}$$

Assuming an average electricity cost of \$0.15/kWh, the yearly operating cost is:

$$C_{\text{server}} = 2190 \times 0.15 = \$328.50/\text{year}$$

$$C_{\text{SBC}} = 438 \times 0.15 = \$65.70/\text{year}$$

$$\Delta C = 328.50 - 65.70 = \$262.80/\text{year saved}$$

This reflects a cost reduction that allows the SBC-based setup to recover its hardware cost within the first year of deployment.

9.2 Environmental Impact

To quantify the environmental implications, carbon emissions were estimated using the internationally recognized global grid average of 0.417 kg CO₂ per kWh.

$$\text{CO}_2^{\text{server}} = 2190 \times 0.417 = 913.2 \text{ kg CO}_2/\text{year}$$

$$\text{CO}_2^{\text{SBC}} = 438 \times 0.417 = 182.5 \text{ kg CO}_2/\text{year}$$

$$\Delta \text{CO}_2 = 913.2 - 182.5 = 730.7 \text{ kg CO}_2/\text{year reduction}$$

The reduction is equivalent to the annual carbon absorption of roughly 30 mature trees or avoiding 3000 km of driving in a petrol vehicle.

10 Future Work

Though the PCB design was complete, hardware testing through it and validating the results with expectations is still left. After a successful trial, it is required to make a cluster combining multiple setups and assess the performance again. Switching from ARM to RISC-V offers Avaota A1 SBC with an octa-core processor (Allwinner T527, 1.8 GHz) and eMMC 5.1 storage. That provides enough power for more enterprise workloads within a similar price range.

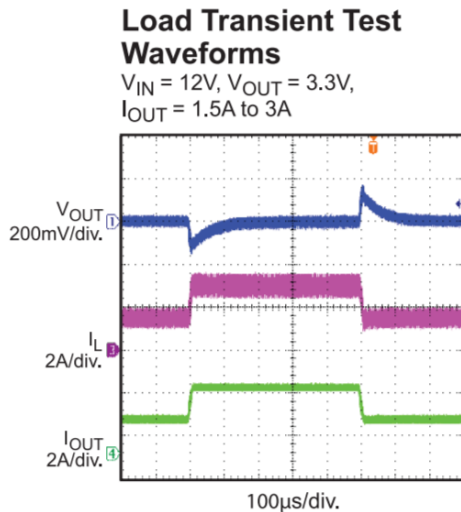


Figure 17: Load balancing of MP1484 [15]

The MP1484 buck module was chosen for its lower fluctuations in voltage with increasing load. It has an exposed pad underneath

for heat dissipation through the PCB. Components listed here may need to be changed for the desired output.

11 Conclusion

This work presents a custom USB RAID controller design that enables ARM SBCs to function as viable server platforms by addressing their fundamental storage expansion limitations. Through strategic selection of dual ASM1352R SATA-to-USB bridge chips paired with an RTS5411T USB hub, the design eliminates SATA port multiplier bottlenecks by providing independent 6 Gbps channels per drive pair. This achieves 2-3x throughput improvements while maintaining superior fault isolation. External power injection via MP1484 buck converters addresses the critical USB power inadequacy. The integration of an STM32-based USB security module provides hardware-secured LUKS key management with read-out protection against physical extraction attempts. The resulting four-layer PCB design, costing approximately \$25 in components, offers configurable RAID modes (0, 1, JBOD, SPAN), making enterprise-grade storage accessible for budget-conscious edge computing deployments. Beyond immediate power savings of up to 59% compared to traditional x86 server solutions, this hardware implementation can influence to set industry standards for server-grade SBC and develop SoCs with dedicated RAID controllers and 10GbE support.

References

- [1] ACASIS. 2023. *Which ACASIS dual-bay hard disk enclosure is best for you?* Retrieved November 21, 2025 from <https://www.acasis.com/blogs/shopping-guide/which-acasis-dual-bay-hard-disk-enclosure-is-best-for-you>
- [2] barryblueice. 2025. *ASM1352R USB Dual SATA Verification*. Retrieved November 21, 2025 from <https://oshwhub.com/barryblueice/asm1352r> Licensed under CERN Open Hardware License.
- [3] Emanuele Cesena. 2019. *Somu: A Tiny FIDO2 Security Key*. Retrieved November 21, 2025 from <https://medium.com/solokeysec/somu-a-tiny-fido2-security-key-5758cc7c142f>
- [4] Realtek Semiconductor Corp. 2021. *RTS5411E - USB3.2 Gen1 4-port Hub Controller*. Retrieved November 21, 2025 from https://www.realtek.com/Product/Index?id=4142&cate_id=801
- [5] Grafick. 2024. *Radxa Zero 3E - Cheapest, Fastest NAS so far? USB3, Gigabit Ethernet, Under \$20*. Retrieved November 21, 2025 from <https://www.youtube.com/watch?v=nwASEvhoO34>
- [6] ASMedia Technology Inc. 2014. *ASM1352R - USB 3.2 Gen2x1 to SATA 6Gbps RAID/Port Multiplier*. Retrieved November 21, 2025 from <https://www.asmedia.com.tw/product/509Yqd4SX5yG9gm7/3AbyQ83xZAUr3qW5>
- [7] The Computer Language Co. Inc. 1998. *Definition: RISC*. Retrieved November 21, 2025 from <https://www.computerlanguage.com/results.php?definition=risc>
- [8] jgreco. 2021. *Multiply your problems with SATA Port Multipliers and cheap SATA controllers*. Retrieved November 21, 2025 from <https://www.truenas.com/community/threads/multiply-your-problems-with-sata-port-multipliers-and-cheap-sata-controllers.96631/>
- [9] JMicon. 2016. *JMB575*. Retrieved November 21, 2025 from <https://www.jmicon.com/products/list/16>
- [10] Jiarong Li. 2015. *"ASMedia ASM1142" Performance Showdown! PCIe 2.0 x2 vs. PCIe 3.0 x1 Interface USB 3.1 Expansion Card Performance Test!* Retrieved November 21, 2025 from <http://www.pcdiy.com.tw/detail/1375>
- [11] Delightly Linux. 2016. *Motherboard RAID or Linux mdadm: Which is Faster?* Retrieved November 21, 2025 from <https://delightlinux.wordpress.com/2016/05/24/motherboard-raid-or-linux-mdadm-which-is-faster/>
- [12] Pedram Kheirkhah Sangdeh and Huacheng Zeng. 2019. Overview of Multiplexing Techniques in Wireless Networks. In *Multiplexing*, S. Mohammady (Ed.). IntechOpen, London, Chapter 3. doi:10.5772/intechopen.85755
- [13] Palanisamy Saravanan. 2020. *SoloKey Full Disk Encryption*. Retrieved November 21, 2025 from <https://github.com/saravanan30erd/solokey-full-disk-encryption>
- [14] superyoshi10000. 2025. *USB Connector*. Retrieved November 21, 2025 from <https://oshwlab.com/superyoshi10000/usb-connector> Licensed under CERN Open Hardware License.
- [15] Monolithic Power Systems. 2008. *MP1484 - 3A, 18V, 340KHz Synchronous Rectified Step-Down Converter Datasheet*. Retrieved November 21, 2025 from <https://www.mouser.com/datasheet/2/277/MP1484-1186021.pdf>